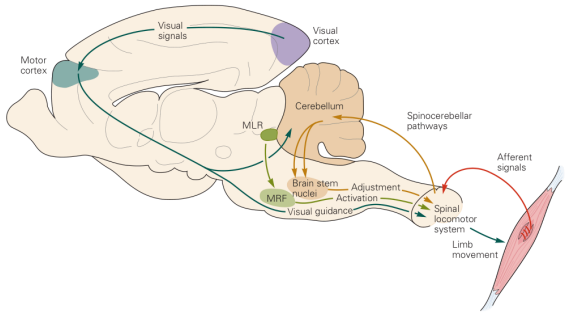
PHYSIOLOGY OF THE MOTOR SYSTEM

Principles of Sensorimotor Control

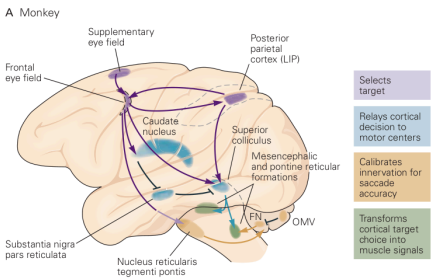
Lecture 01

Different motor tasks, different circuit layouts

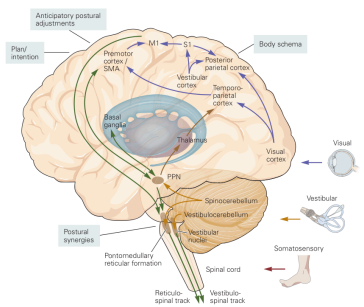
LOCOMOTION



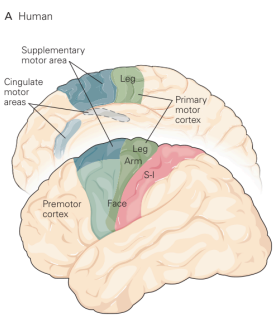
GAZE



POSTURE



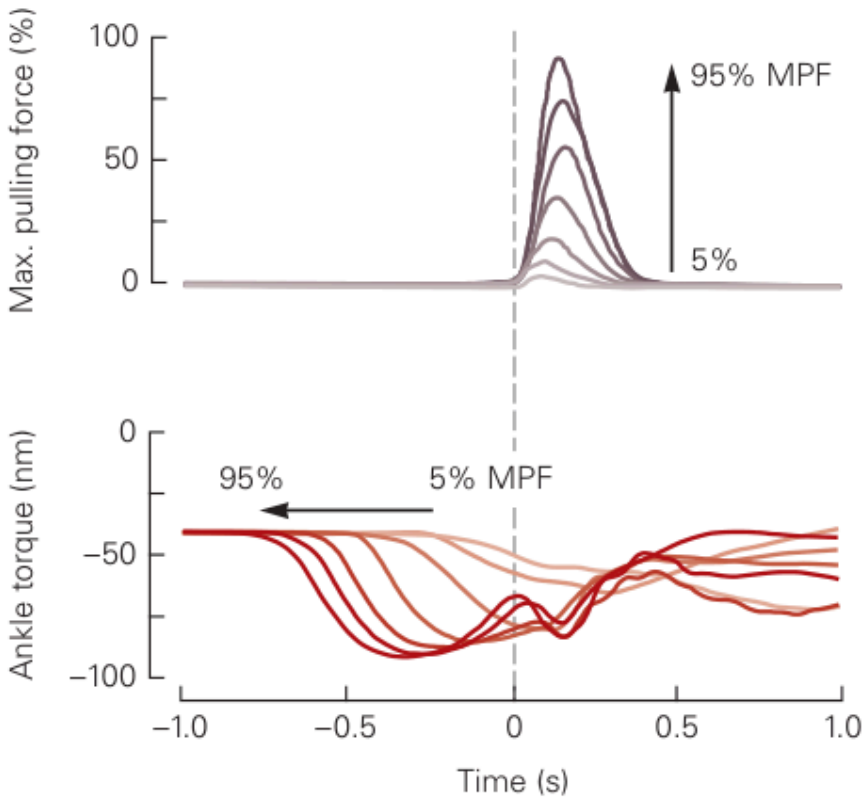
VOLUNTARY ACTION



> These tasks draw on differently organized, interconnected circuits.

Kandel 6e, Figs. 33–1, 35–9A, 36–13, 34–2A · selective circuit maps.

Does ankle torque change before the arm pull?

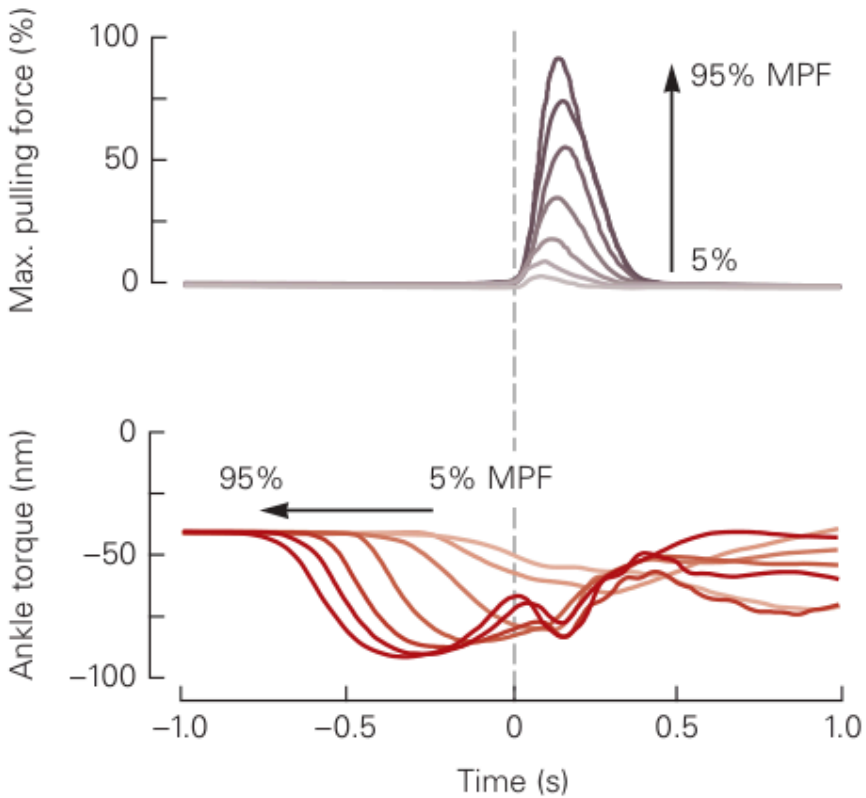


Kandel 6e, Fig. 36–8A, p. 893 · arm force and ankle torque

READ THE TRACES

Time zero: pulling force begins at the handle. When does ankle torque change?

Does ankle torque change before the arm pull?



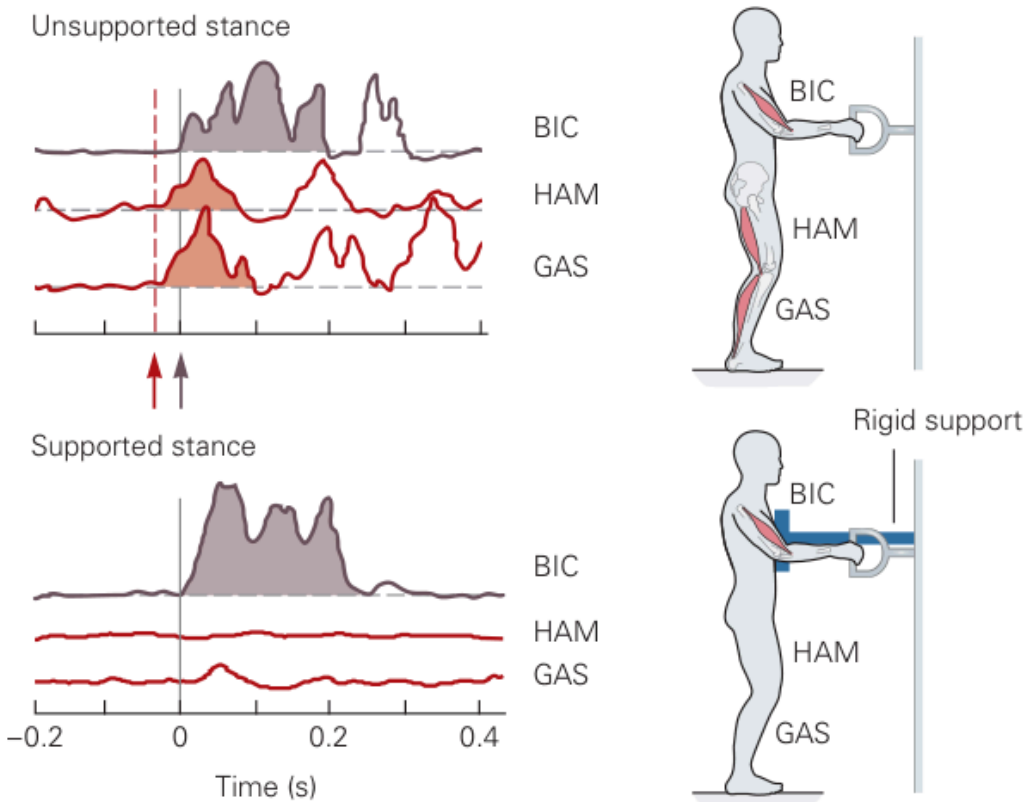
Kandel 6e, Fig. 36–8A, p. 893 · arm force and ankle torque

READ THE TRACES

Time zero: pulling force begins at the handle. When does ankle torque change?

> Ankle torque changes before the measured arm pull.

Does external support change postural recruitment?

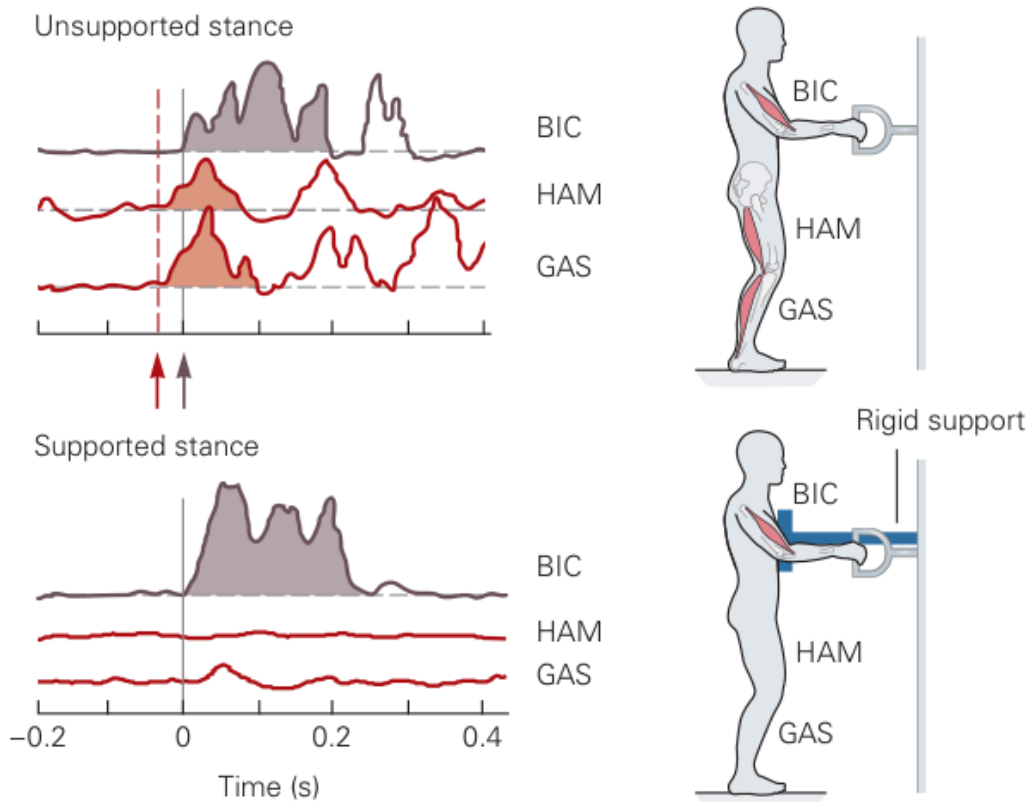


Kandel 6e, Fig. 36–8B, p. 893 · EMG and support manipulation

COMPARE CONDITIONS

Time zero: biceps activation. Compare leg muscle traces with and without rigid support.

Does external support change postural recruitment?



Kandel 6e, Fig. 36–8B, p. 893 · EMG and support manipulation

COMPARE CONDITIONS

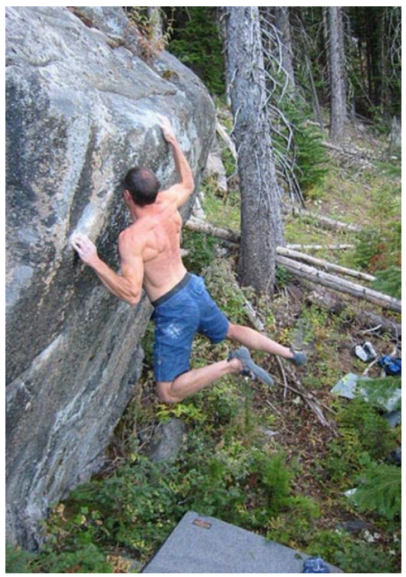
Time zero: biceps activation. Compare leg muscle traces with and without rigid support.

> Hamstring and gastrocnemius recruitment is reduced with support.

Voluntary, rhythmic, and reflexive are control modes

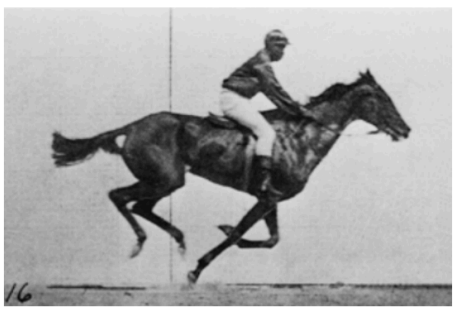
Movement classes: overlapping levels of control

VOLUNTARY



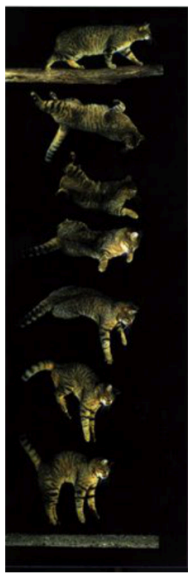
Flexible goal-directed actions

RHYTHMIC



Centrally generated sequences

REFLEXIVE



Stimulus-linked responses

Kandel 6e, Ch. 30, pp. 715–716; adapted prior L01 examples · conceptual illustration

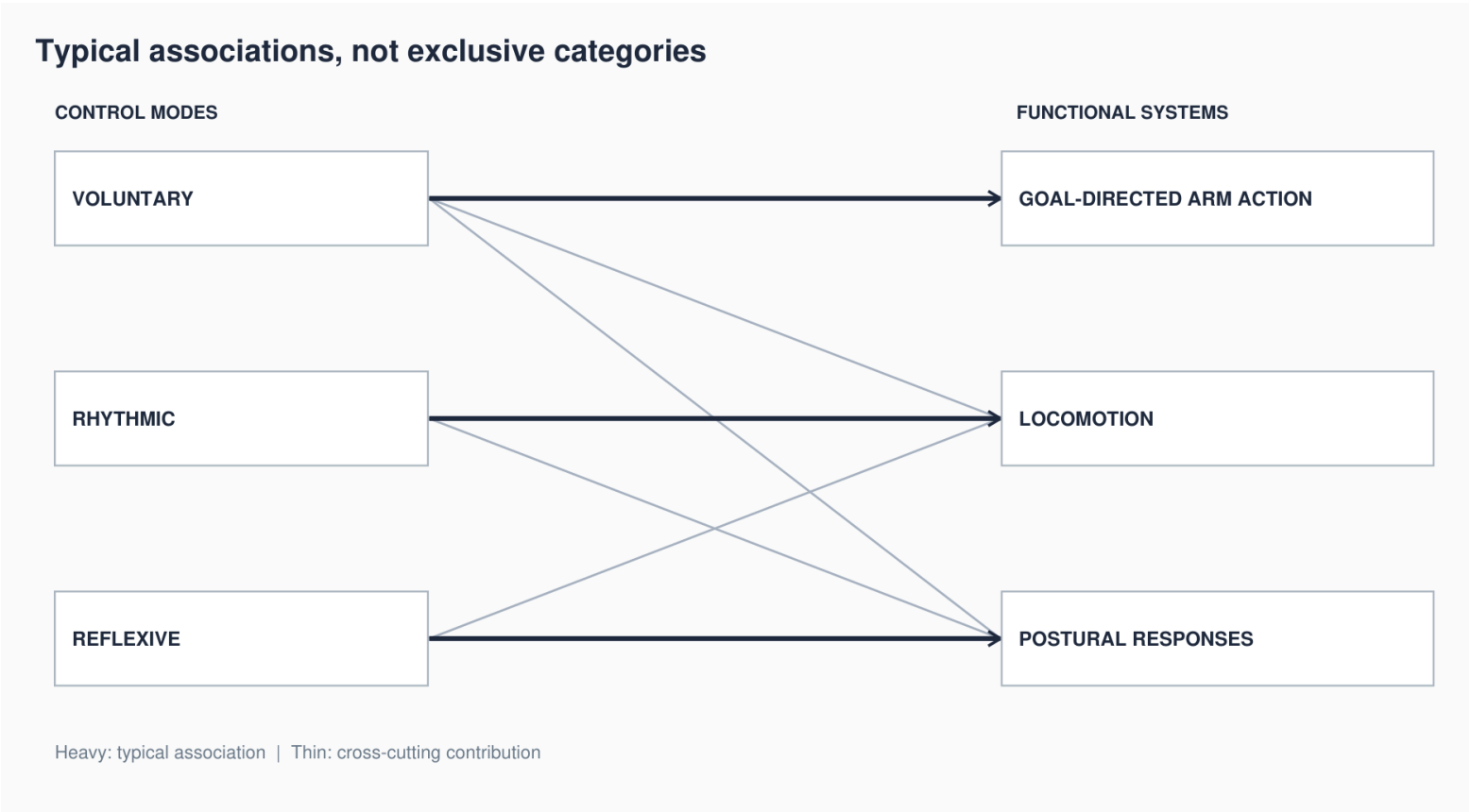
The modes differ in typical control demands

VOLUNTARY	RHYTHMIC	REFLEXIVE
Flexible, goal-dependent initiation	Recurrent patterns with ongoing control	Relatively rapid, stimulus-linked response

> The same action may combine all three modes.

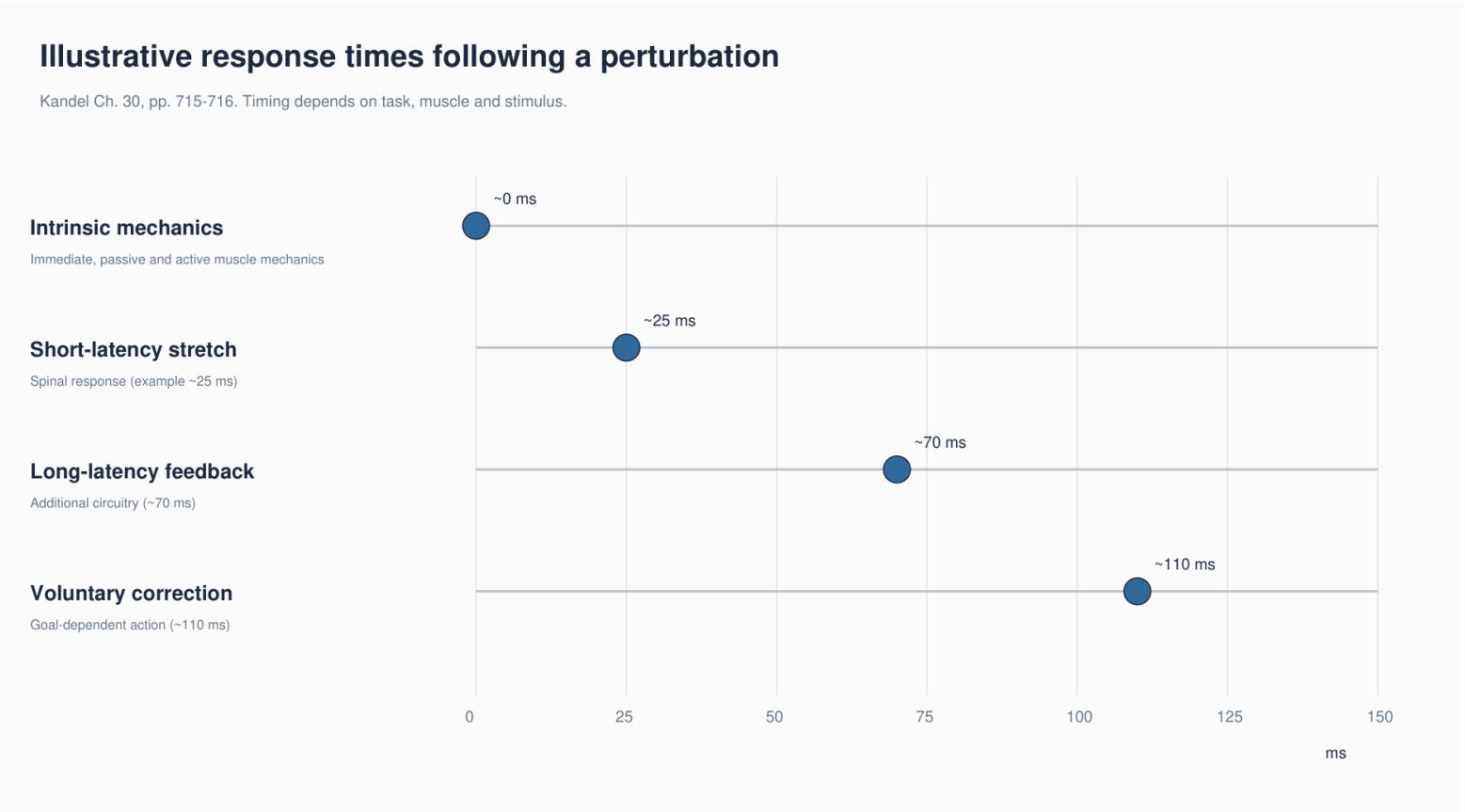
Kandel 6e, Ch. 30, pp. 715–716 · teaching synthesis.

Modes cross the functional systems



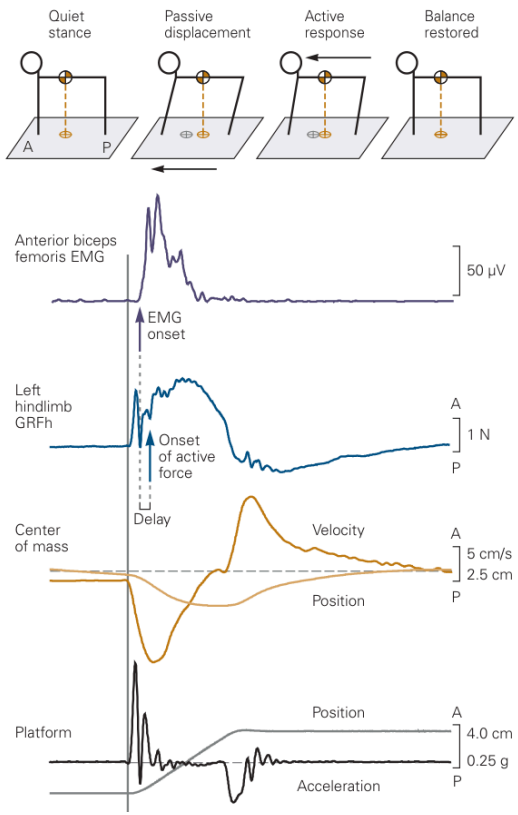
Kandel 6e, Ch. 30 and systems chapters · instructor–specialist conceptual map

Mechanics and neural responses occupy different timescales



Kandel 6e, Ch. 30, pp. 715–716 · illustrative latencies

A postural perturbation produces several measured responses

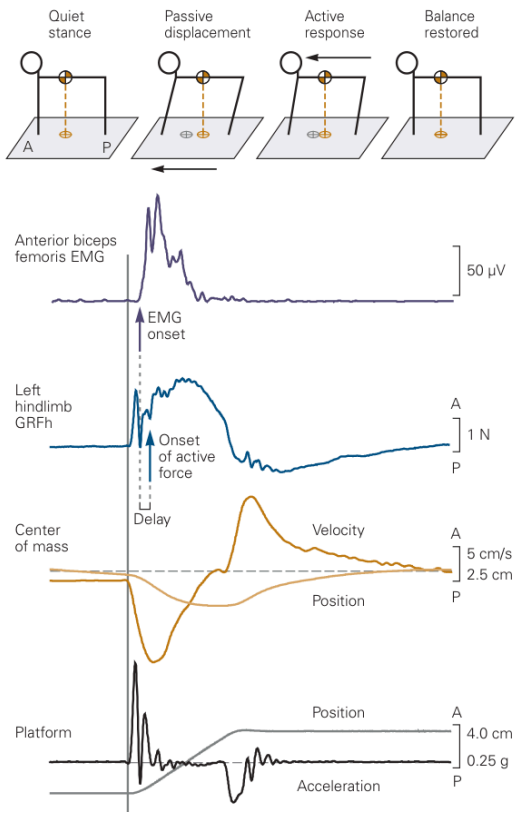


Kandel 6e, Fig. 36–3, p. 888 · EMG and force recordings

READ DOWN THE TRACES

Platform motion, muscle activity, center of mass, and corrective motion unfold over time.

A postural perturbation produces several measured responses



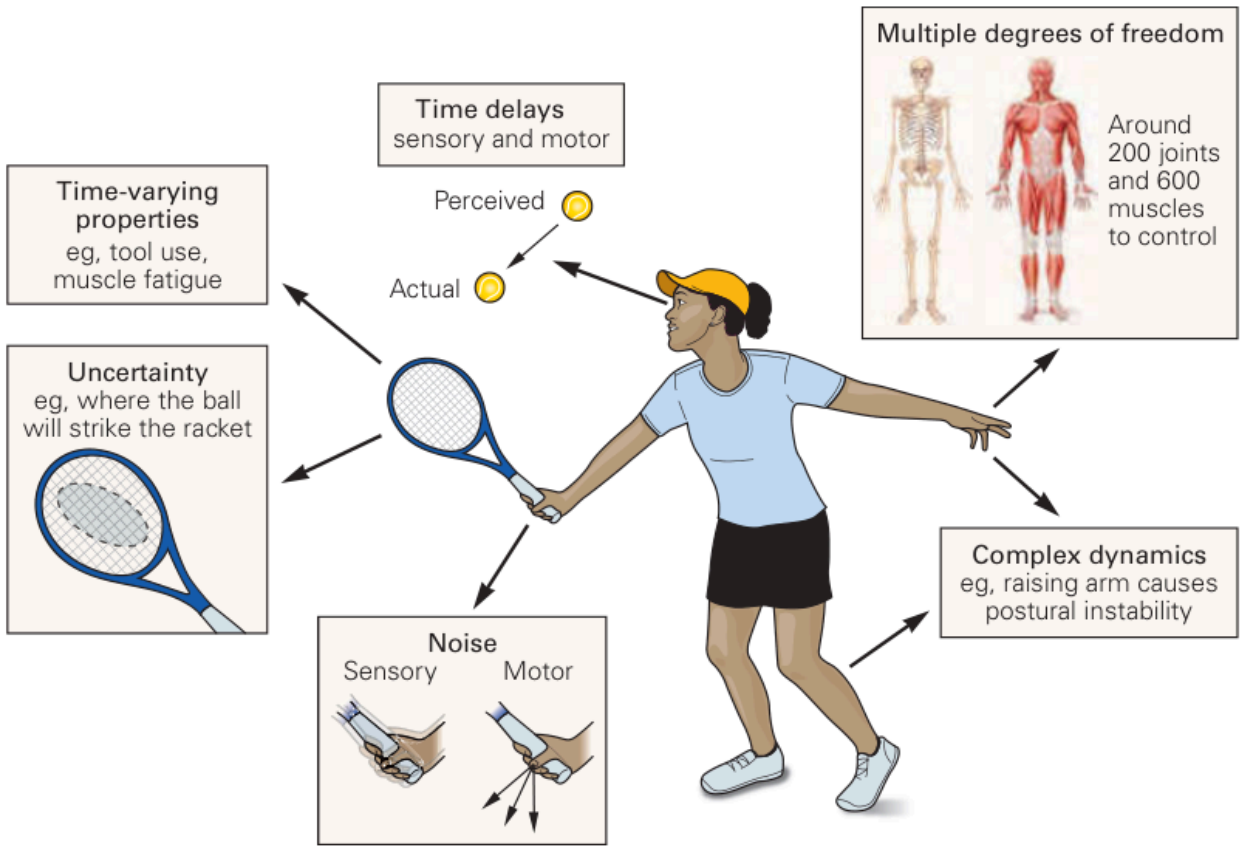
Kandel 6e, Fig. 36–3, p. 888 · EMG and force recordings

READ DOWN THE TRACES

Platform motion, muscle activity, center of mass, and corrective motion unfold over time.

> The measured response is a coupled mechanical and neural process.

One movement faces several control challenges



Kandel 6e, Fig. 30–1, p. 714 · schematic illustration

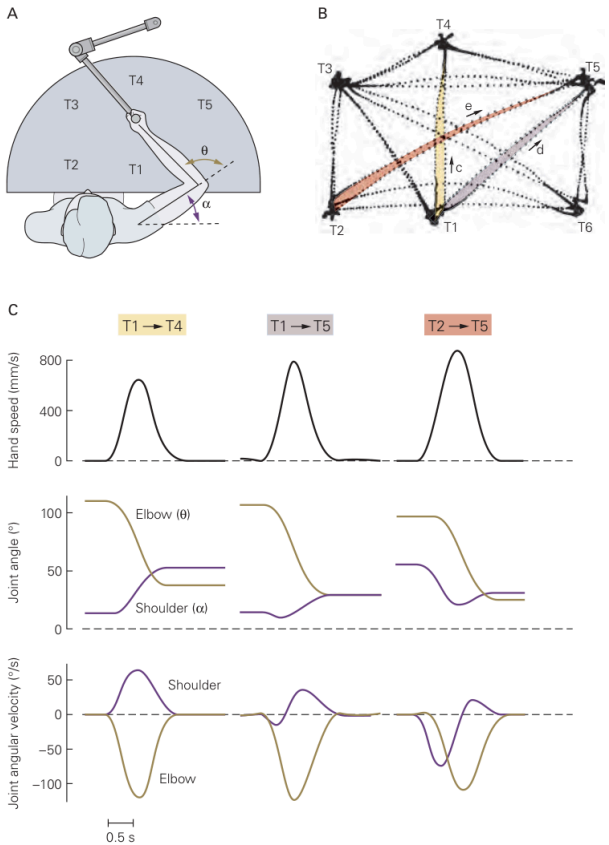
Many joints and muscles serve a few goals

BEHAVIORAL GOAL	CONTROL VARIABLES
Place the hand at the target	Coordinate joints and muscles whose effects are coupled

> The controller must manage many possible solutions under uncertain, delayed feedback.

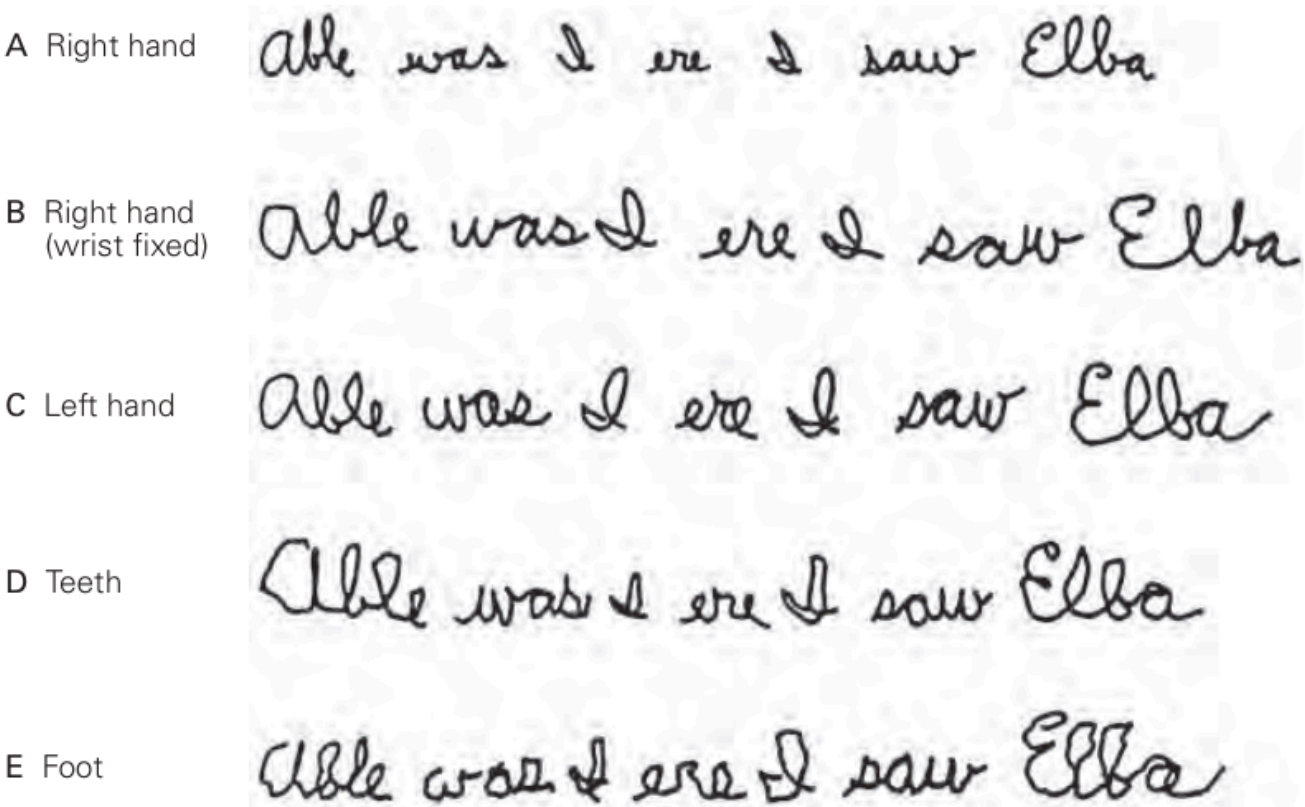
Kandel 6e, Fig. 30–1, pp. 714–715 · conceptual synthesis.

Similar hand paths can arise from complex joint motion



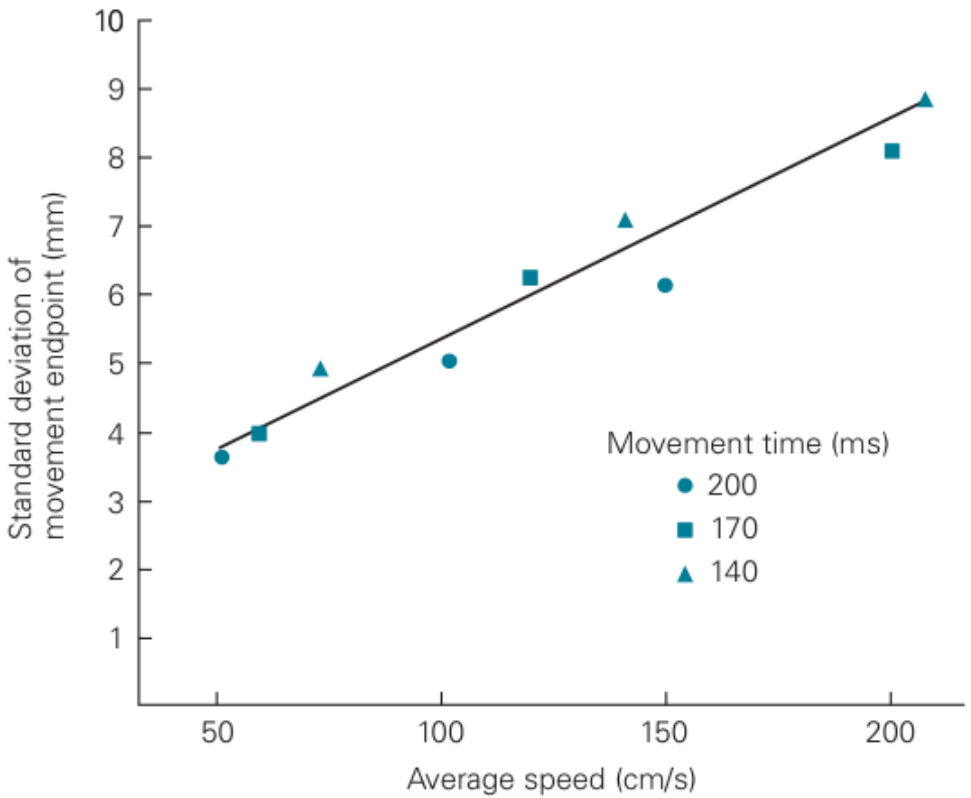
Kandel 6e, Fig. 30–8A–C, p. 726 · reaching kinematics

A recognizable letter survives a change of effector



Kandel 6e, Fig. 30–9A–E, p. 727 · handwriting observation

Faster reaches show more endpoint variability

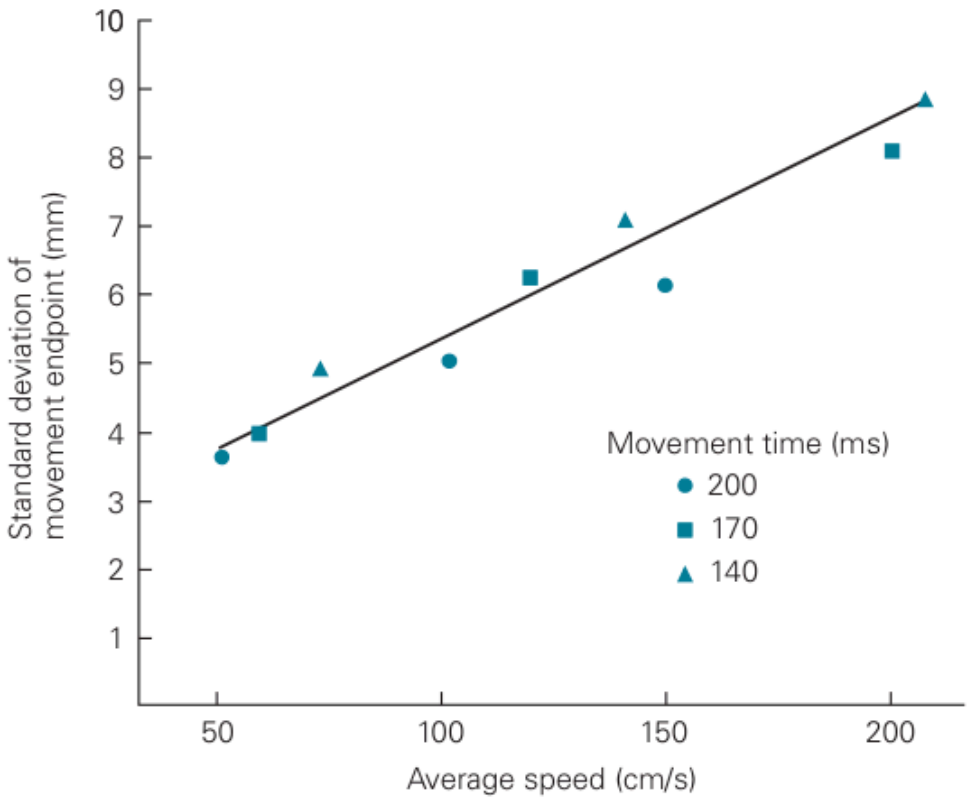


Kandel 6e, Fig. 30–10, p. 728 · endpoint SD vs speed

READ THE Y AXIS

This is variability, not mean directional bias.

Faster reaches show more endpoint variability



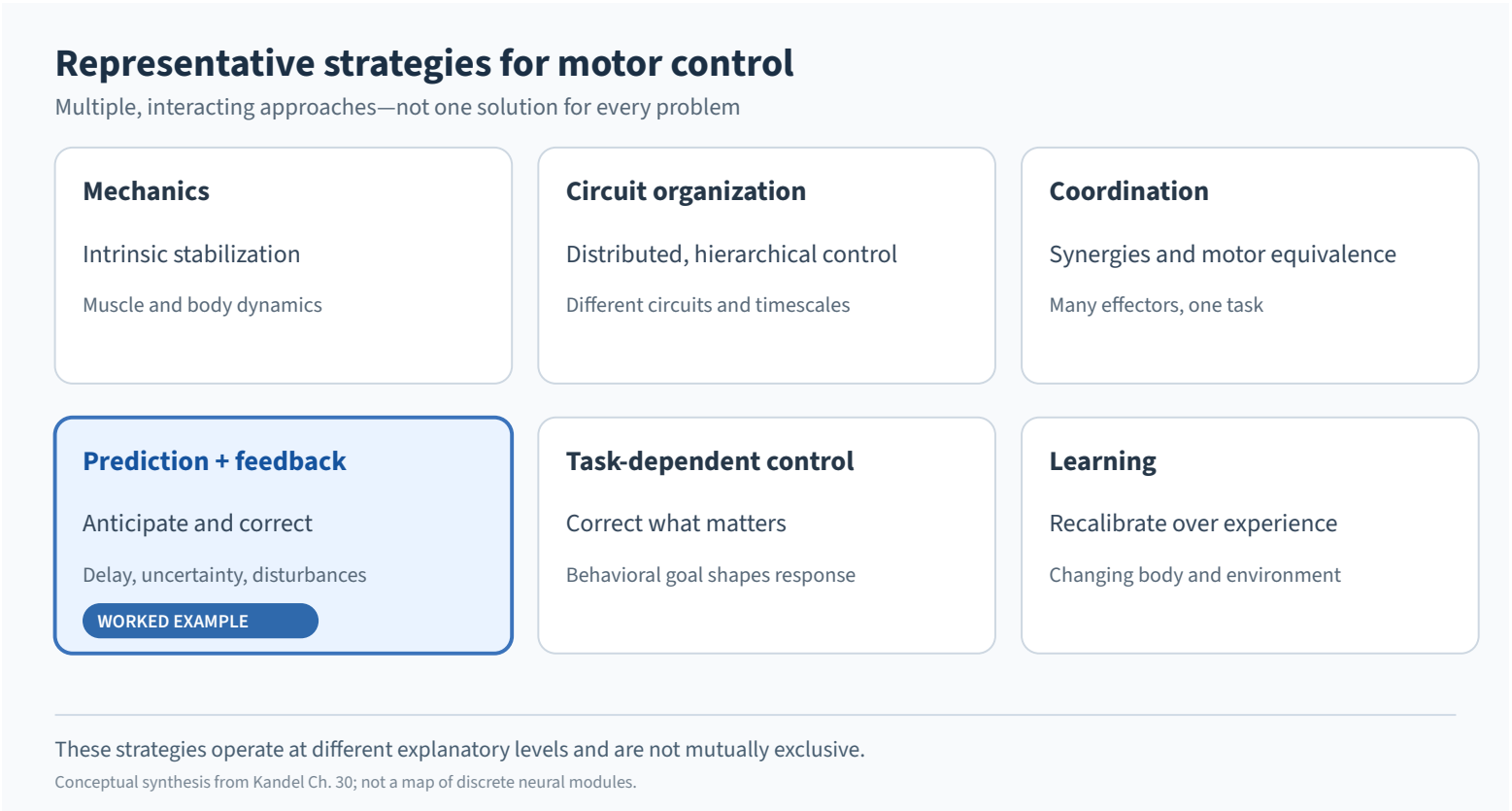
Kandel 6e, Fig. 30–10, p. 728 · endpoint SD vs speed

READ THE Y AXIS

This is variability, not mean directional bias.

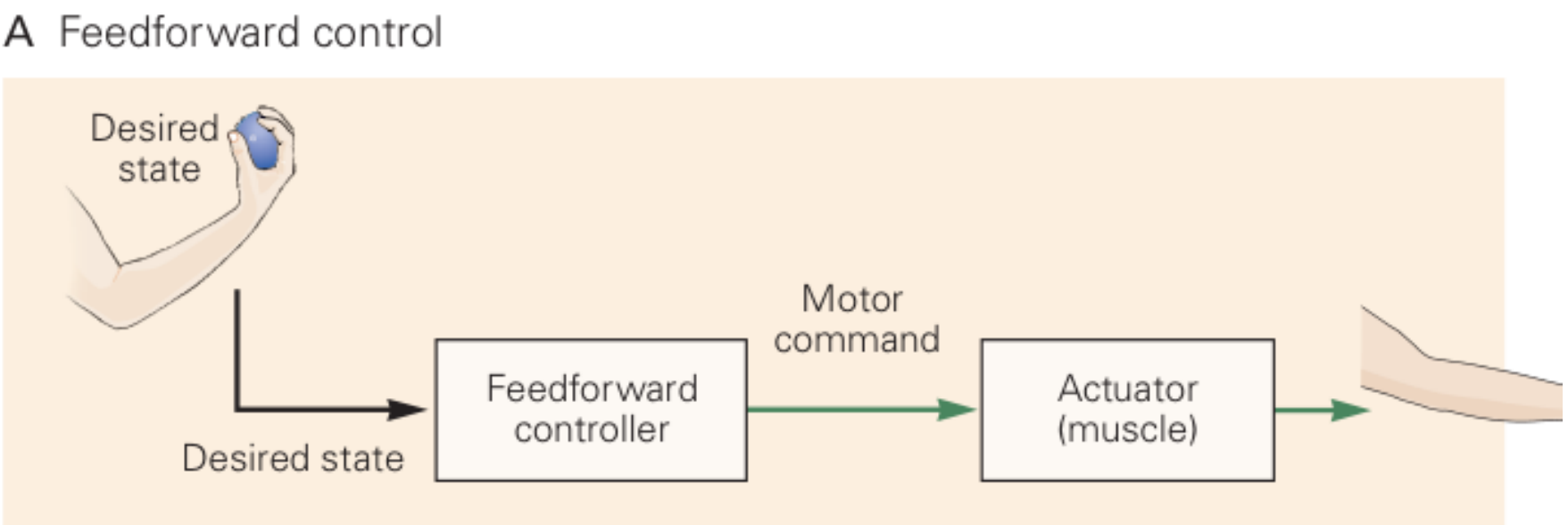
> Speed and precision trade off in these data.

Motor systems use complementary strategies



Kandel 6e, Ch. 30 · conceptual synthesis, not separate brain modules

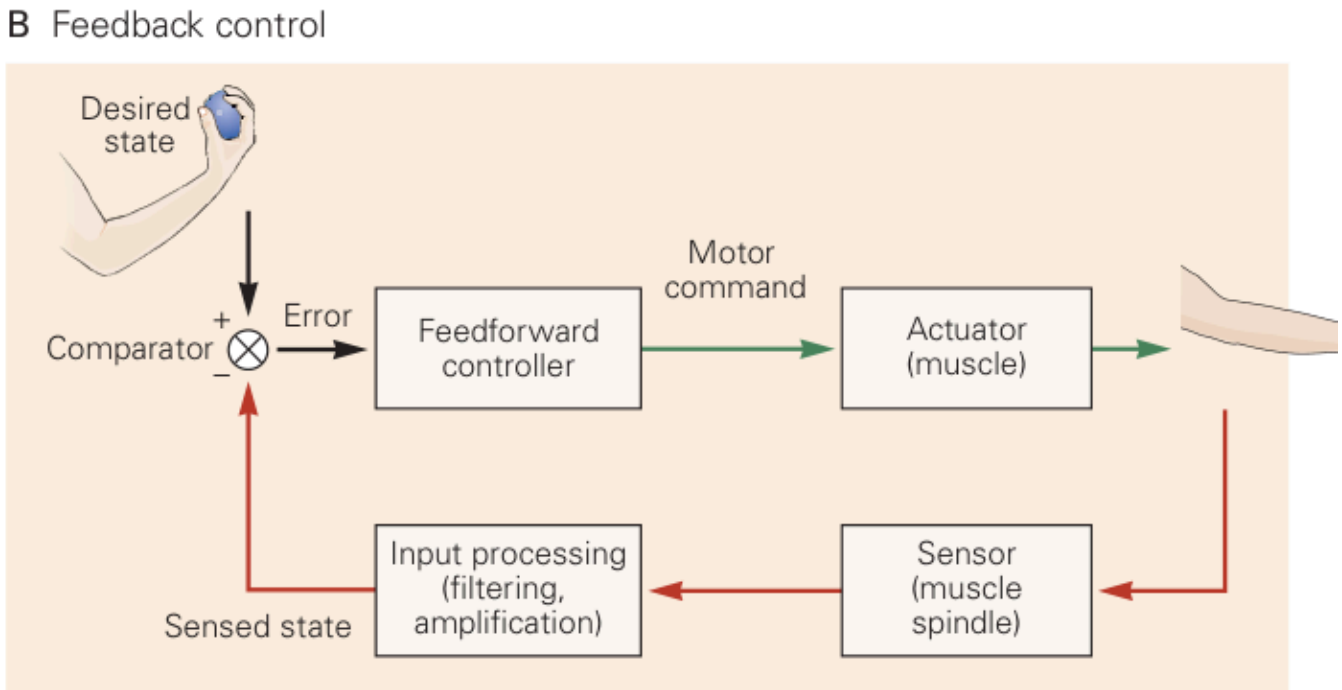
Feedforward commands act before new sensory correction



Kandel 6e, Fig. 30–2A, p. 717 · model diagram

> Advance control can be fast, but an unforeseen disturbance is not corrected by this pathway alone.

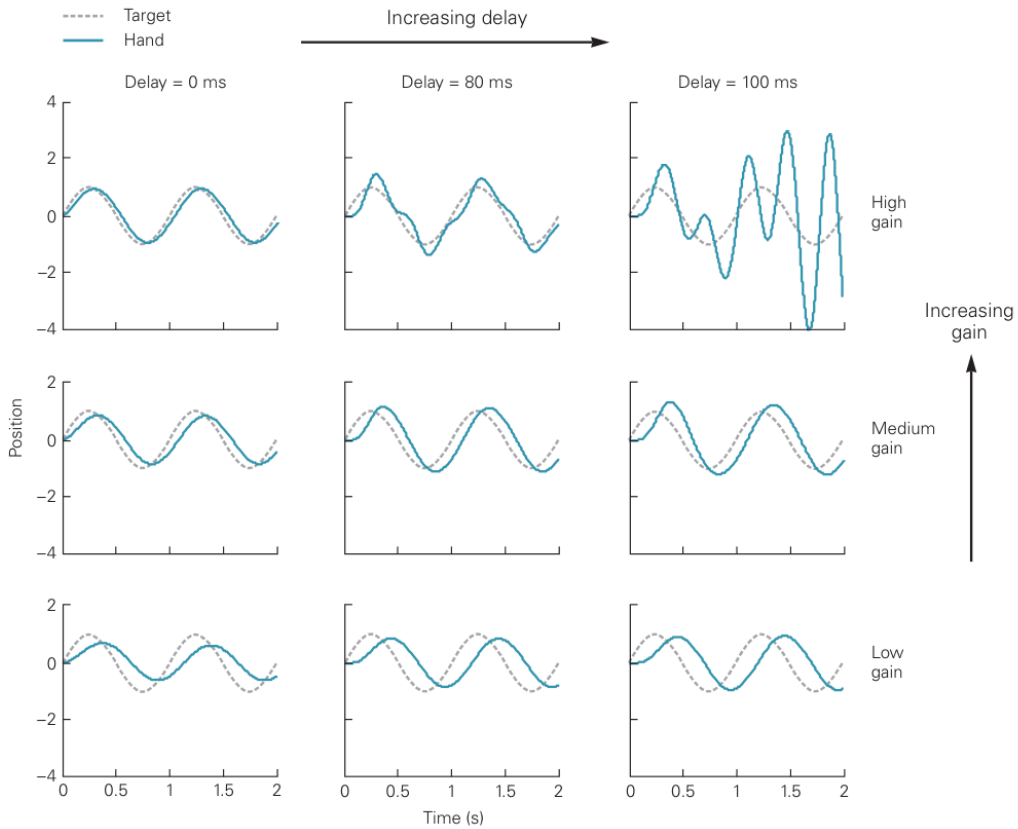
Feedback compares an outcome with a desired state



Kandel 6e, Fig. 30–2B, p. 717 · model diagram

> Online correction can address disturbances, but its sensory estimate arrives with delay.

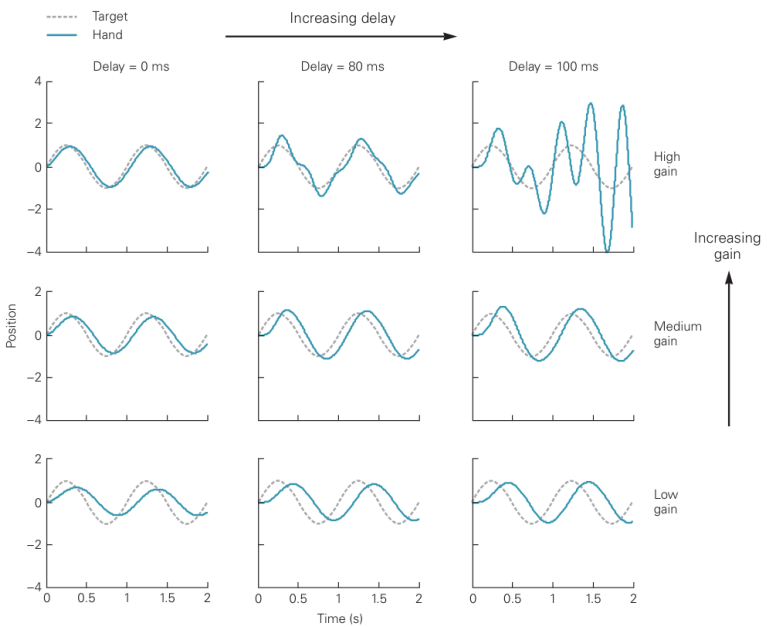
Gain and delay jointly determine tracking



Kandel 6e, Fig. 30–4, p. 720 · controller simulation

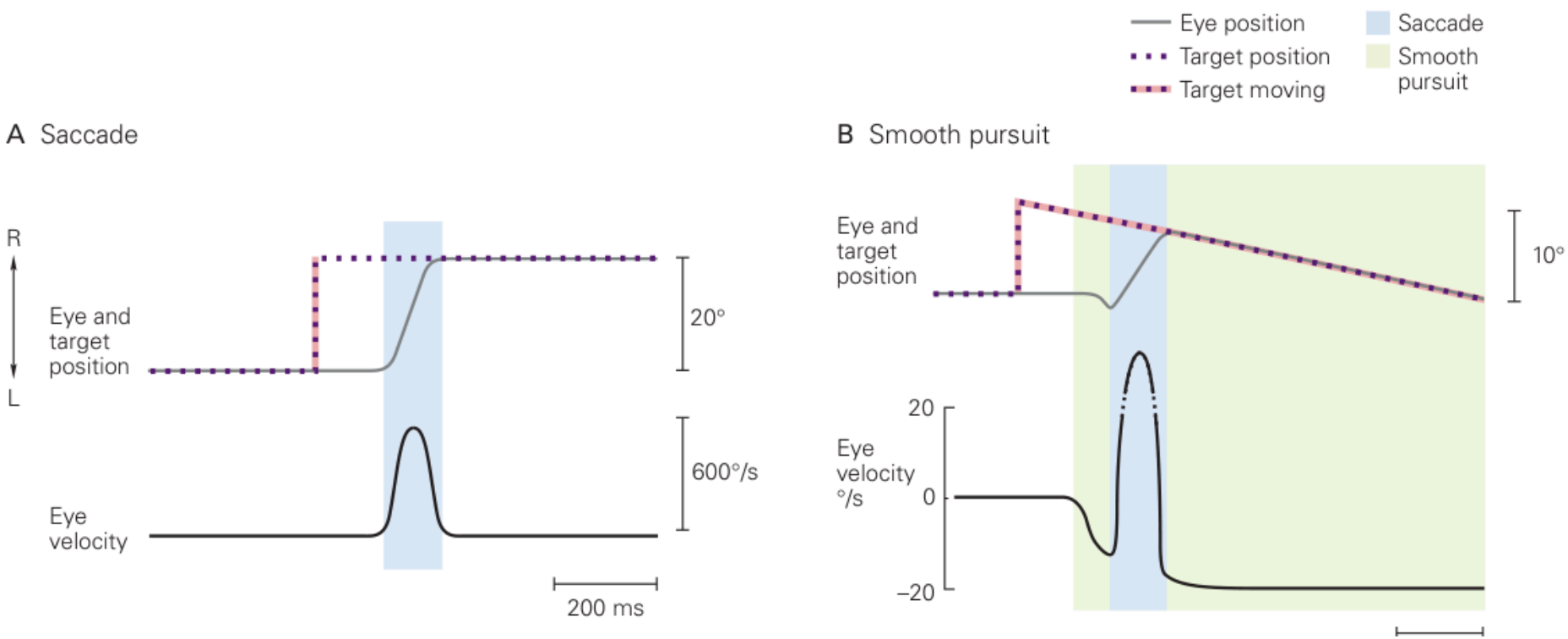
Strong delayed correction can oscillate

> More correction is not always better when the signal describes an earlier state.



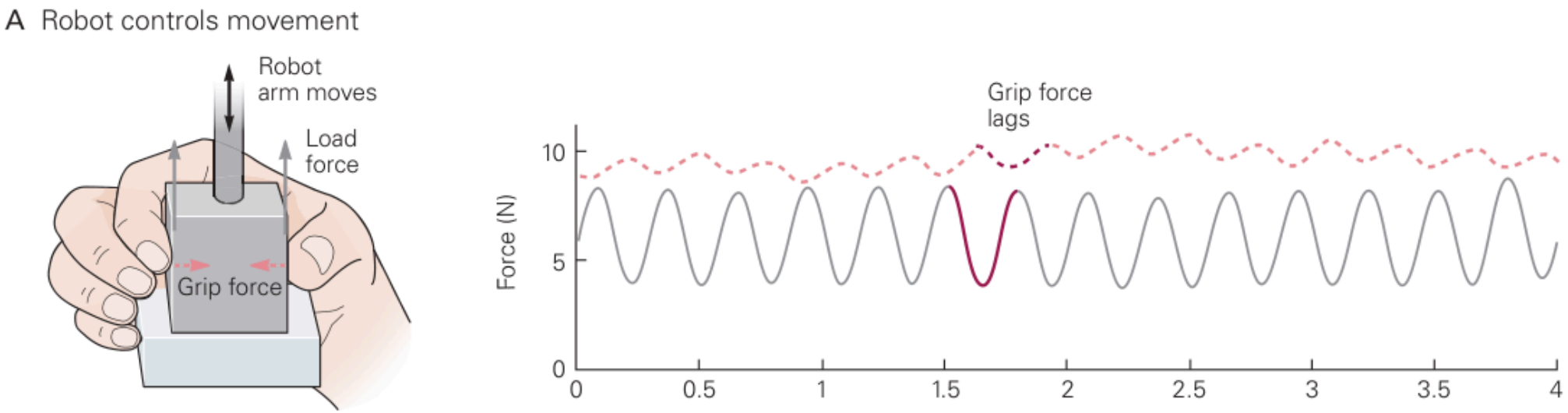
Kandel 6e, Fig. 30–4, p. 720 · simulation, not neural recording

Gaze uses control strategies on different timescales



Kandel 6e, Fig. 35–6A/B, p. 867 · eye movement recordings

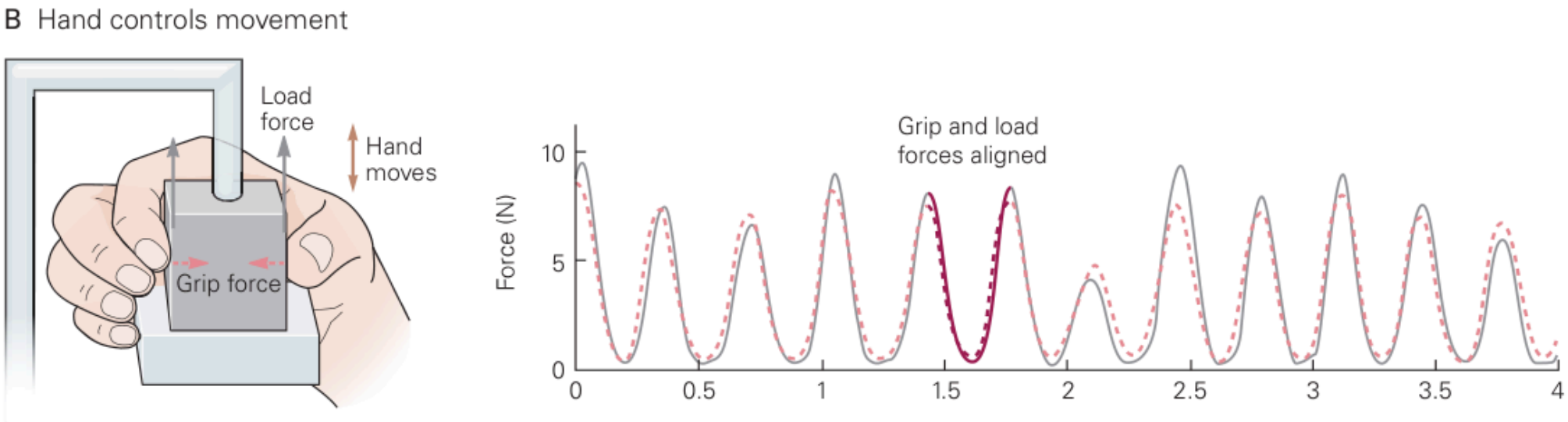
External loading elicits a delayed grip response



Kandel 6e, Fig. 30–6A, p. 724 · externally produced load

PINK DASHED · GRIP FORCE | GRAY SOLID · LOAD FORCE

Self-generated loading aligns grip and load force

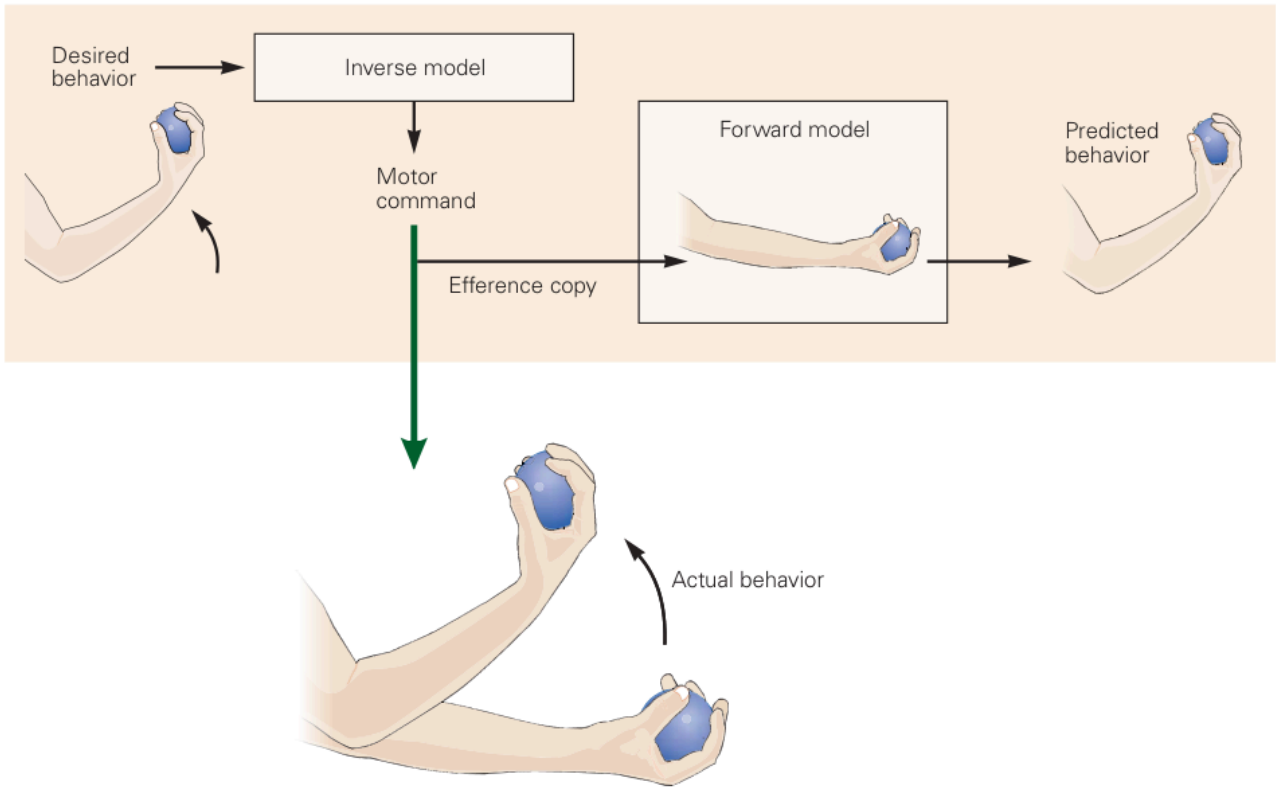


Kandel 6e, Fig. 30–6B, p. 724 · self-produced load

PINK DASHED · GRIP FORCE | GRAY SOLID · LOAD FORCE

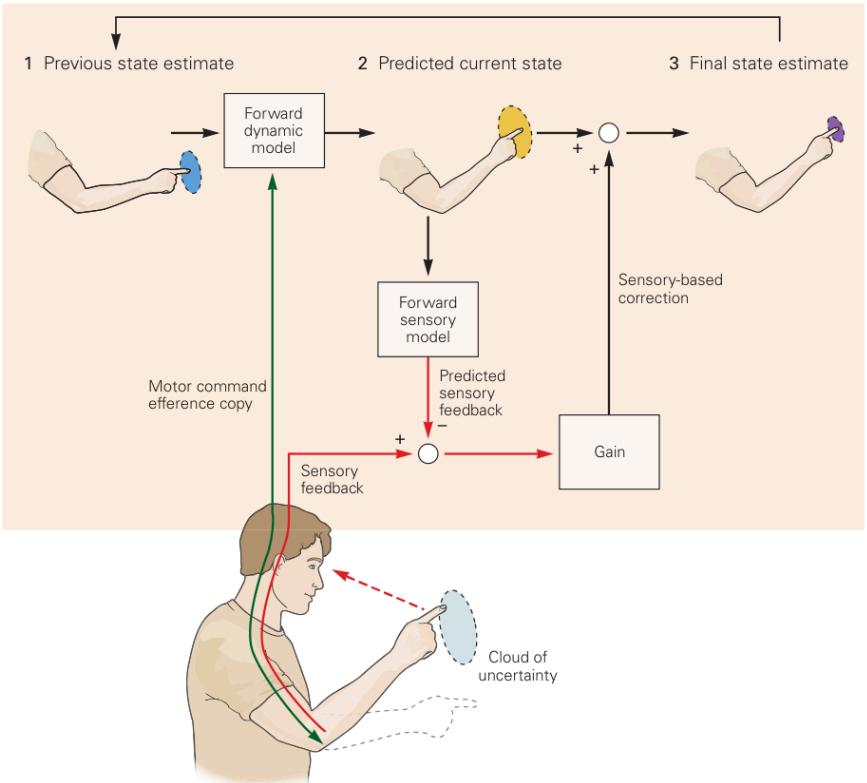
> The closer phase match supports anticipatory control.

Internal models propose distinct computations



Kandel 6e, Fig. 30–3, p. 718 · forward and inverse model schematic

An observer combines prediction with sensation



Kandel 6e, Fig. 30–5, p. 722 · state-observer model

State estimation addresses delayed, noisy feedback

PREDICTION

Use the previous estimate and a copy of the command

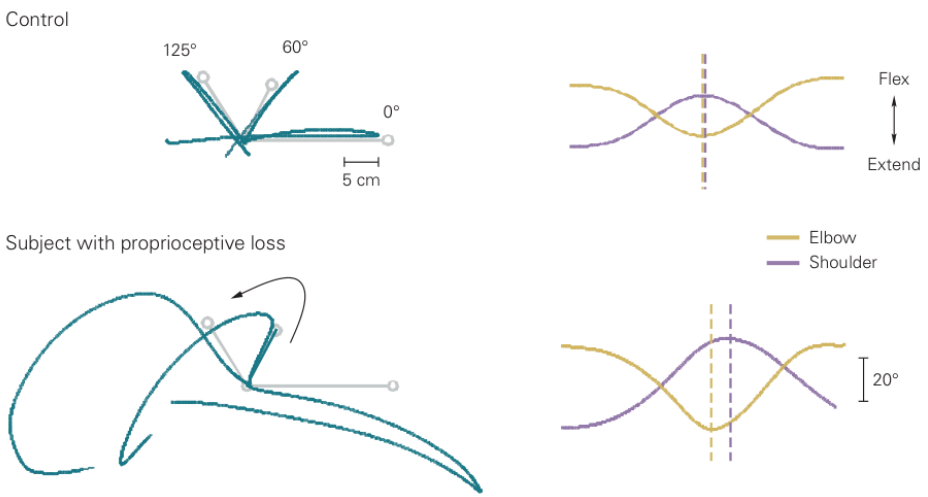
MEASUREMENT

Use incoming sensory evidence to correct the estimate

> Neither source alone is sufficient across all conditions.

Kandel 6e, Figs. 30–3 and 30–5, pp. 718, 722 · model synthesis.

Loss of proprioception disrupts rapid reversal

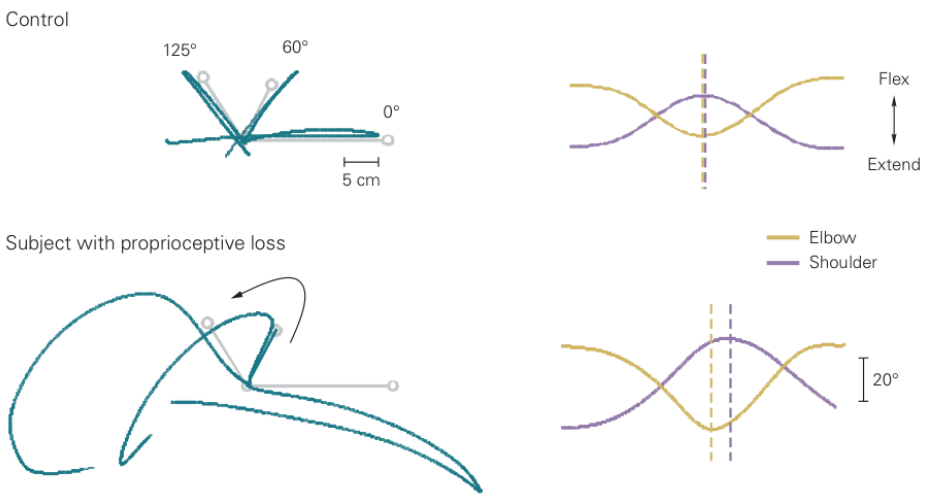


COMPARE THE TRACES

How do limb paths and joint timing differ at the reversal?

Kandel 6e, Fig. 30–14, p. 733 · control and deafferented patient

Loss of proprioception disrupts rapid reversal



COMPARE THE TRACES

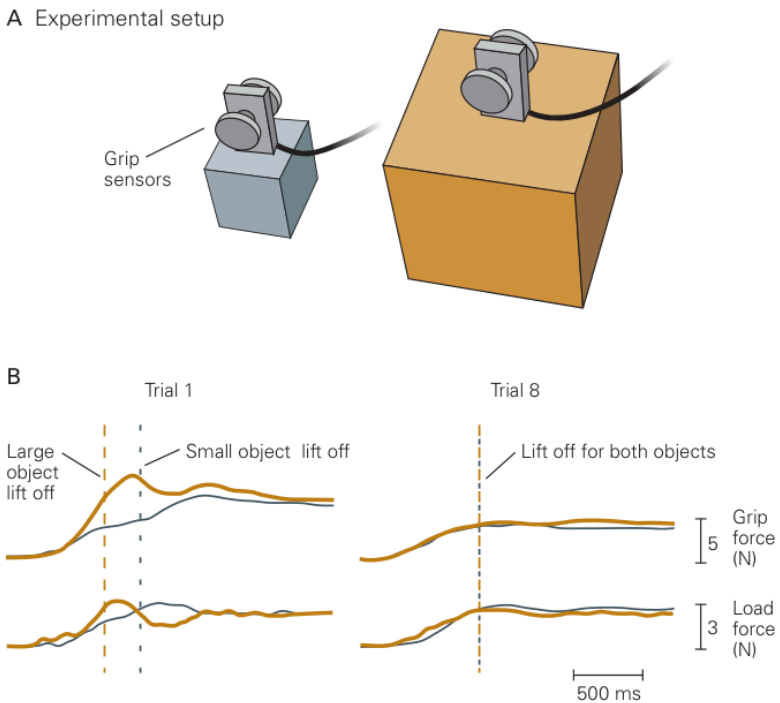
How do limb paths and joint timing differ at the reversal?

> Usable state information matters for multijoint coordination.

Kandel 6e, Fig. 30–14, p. 733 · control and deafferented patient

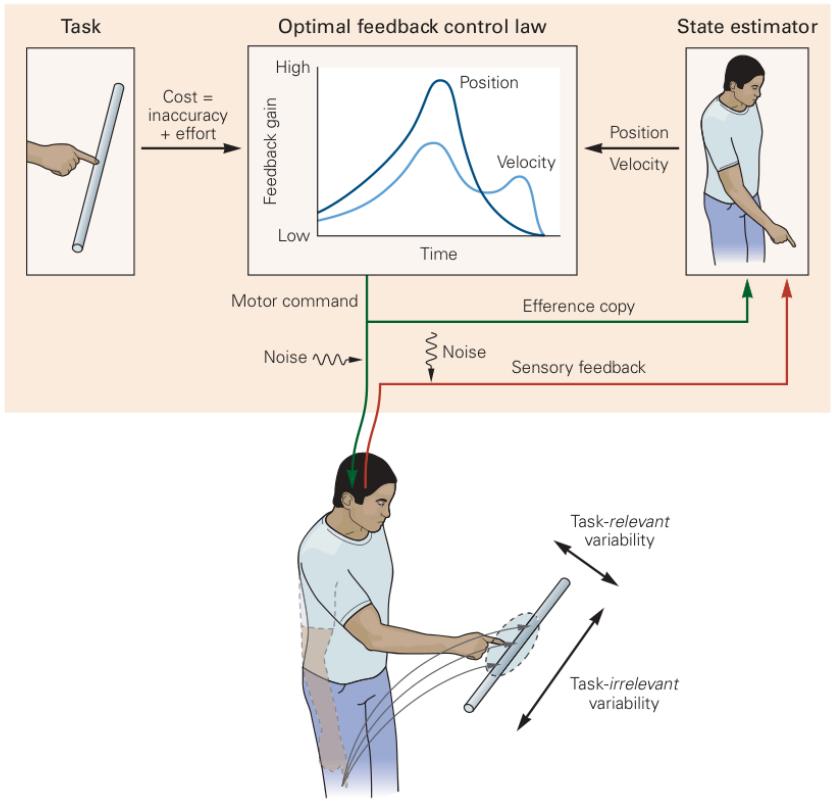
Grip scaling changes while heaviness perception persists

EQUAL WEIGHT · ORANGE = LARGE · GRAY = SMALL



Kandel 6e, Fig. 30–7A/B, p. 725 · size–weight illusion experiment

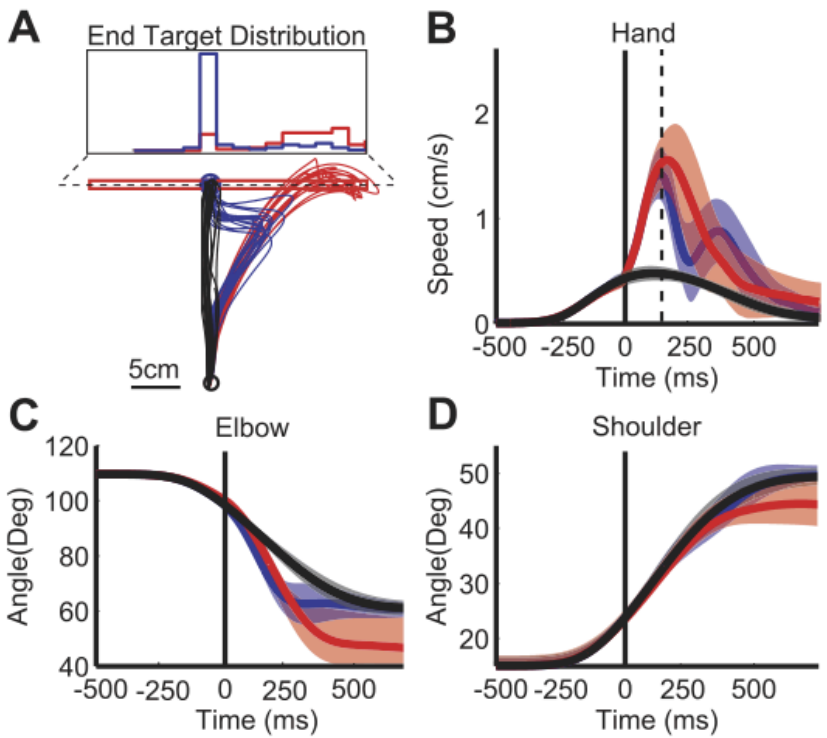
Task goals determine which deviations need correction



Kandel 6e, Fig. 30–11, p. 729 · optimal feedback control model

A dot and a bar set different correction demands

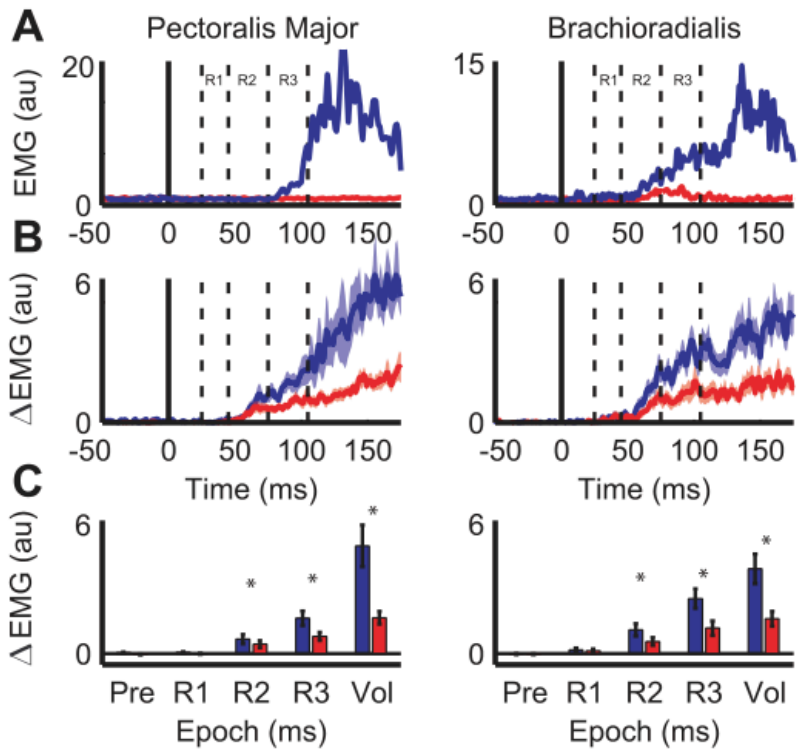
BLUE · DOT TARGET | RED · BAR TARGET



Nashed et al. 2012, Fig. 2A–D, p. 1003 · perturbation kinematics

Later muscle responses reflect task relevance

BLUE · DOT TARGET | RED · BAR TARGET



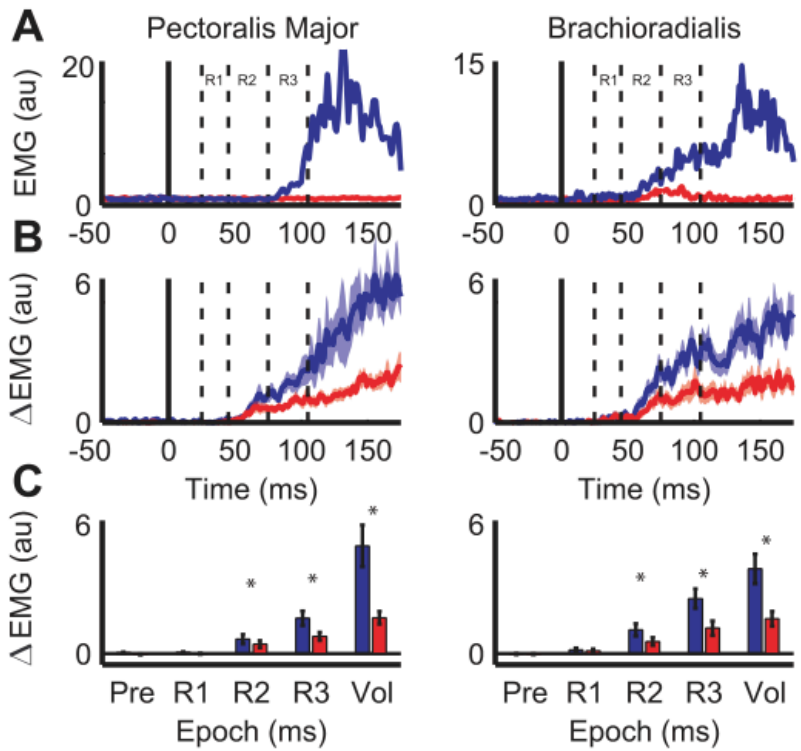
Nashed et al. 2012, Fig. 3A–C, p. 1003 · EMG

COMPARE EPOCHS

Is the early R1 response or the later R2/R3 response more task sensitive?

Later muscle responses reflect task relevance

BLUE · DOT TARGET | RED · BAR TARGET



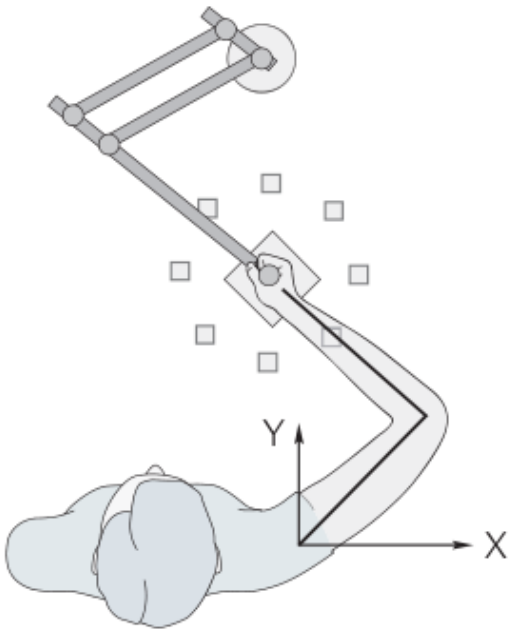
Nashed et al. 2012, Fig. 3A–C, p. 1003 · EMG

COMPARE EPOCHS

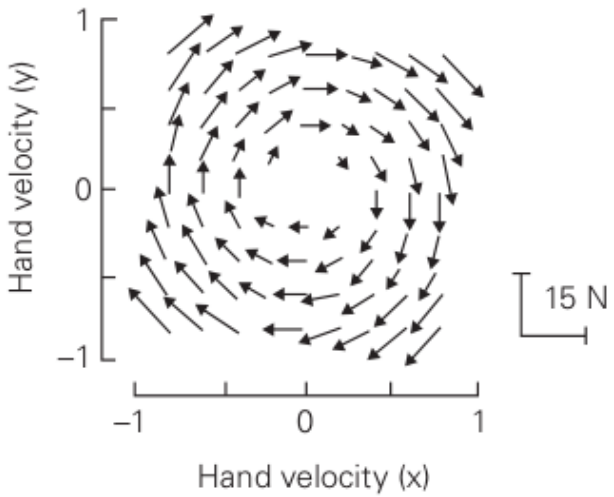
Is the early R1 response or the later R2/R3 response more task sensitive?

> Later perturbation responses differ with target geometry.

A robotic handle applies a velocity-dependent force



Kandel 6e, Fig. 30–12A, p. 731 · apparatus

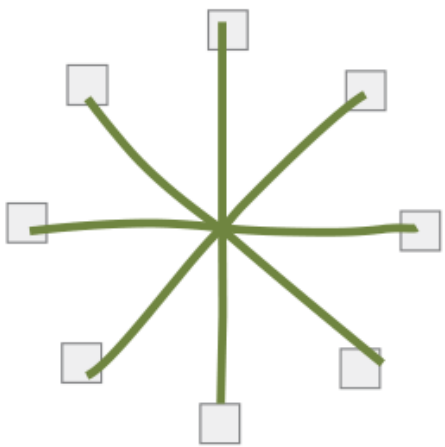


Kandel 6e, Fig. 30–12C, p. 731 · applied force field

The device perturbs the hand during reaches to different targets.

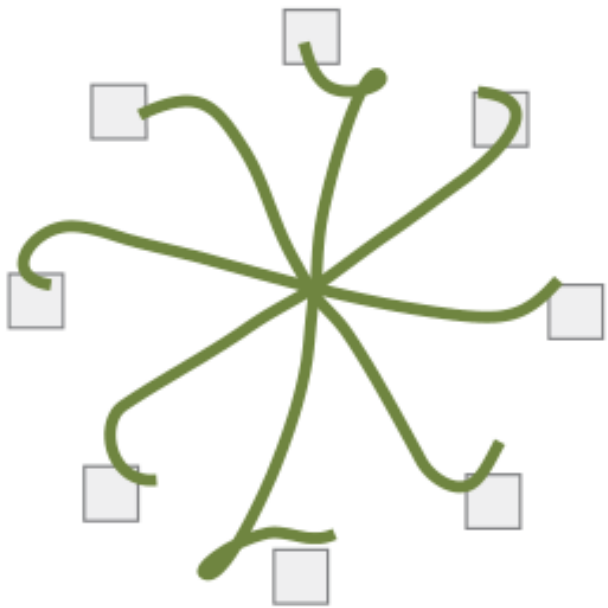
What changes when the force field turns on?

NULL FIELD



Kandel 6e, Fig. 30–12B, p. 731 · null-field paths

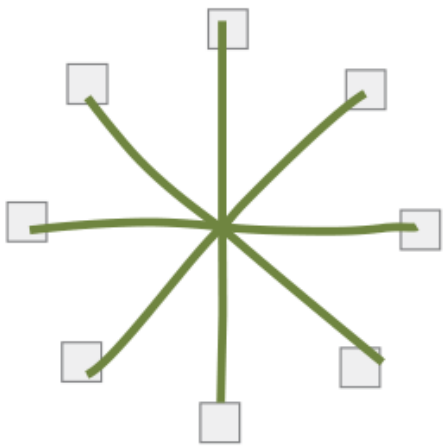
FIELD ON · FIRST REACHES



Kandel 6e, Fig. 30–12D1, p. 731 · initial paths

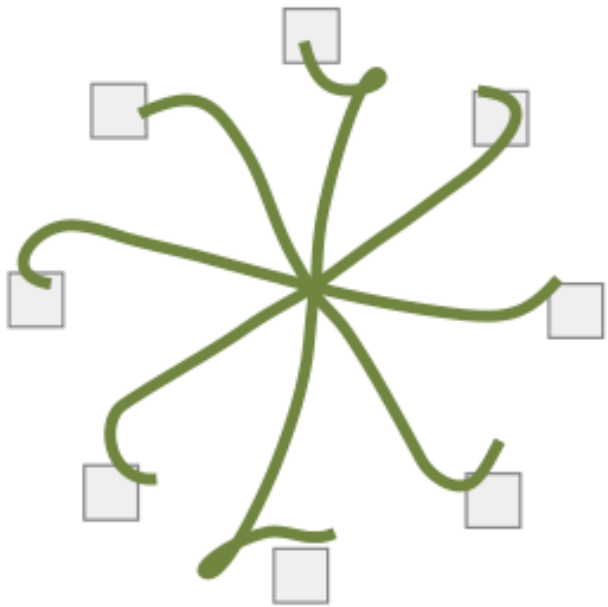
What changes when the force field turns on?

NULL FIELD



Kandel 6e, Fig. 30–12B, p. 731 · null-field paths

FIELD ON · FIRST REACHES

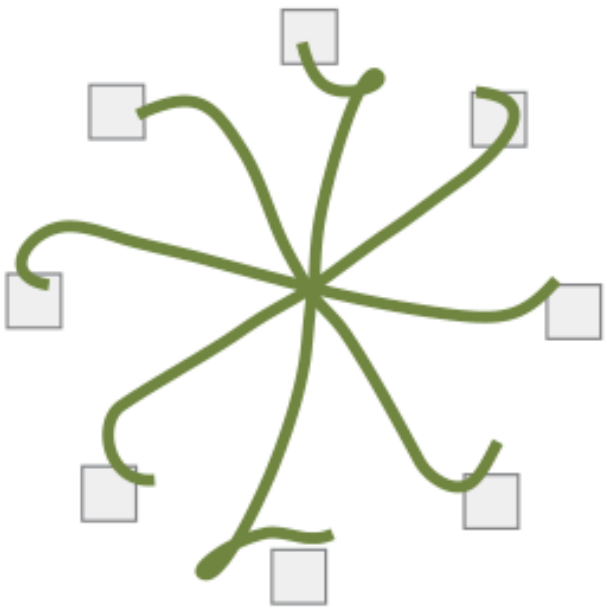


Kandel 6e, Fig. 30–12D1, p. 731 · initial paths

With the force field on, initial paths curve away from the null-field paths.

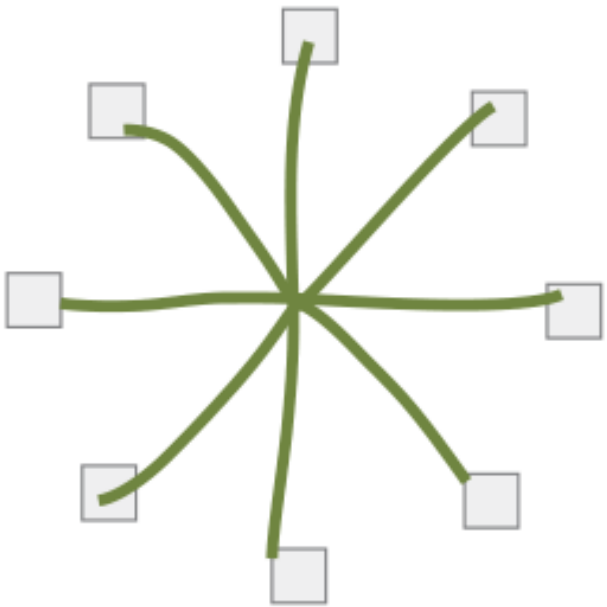
What changes with practice in the field?

FIELD ON · FIRST REACHES



Kandel 6e, Fig. 30–12D1, p. 731 · first reaches

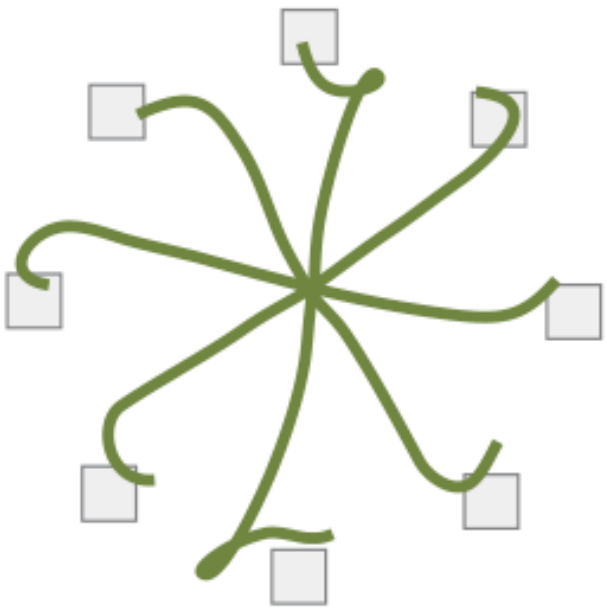
FIELD ON · AFTER PRACTICE



Kandel 6e, Fig. 30–12D2, p. 731 · later reaches

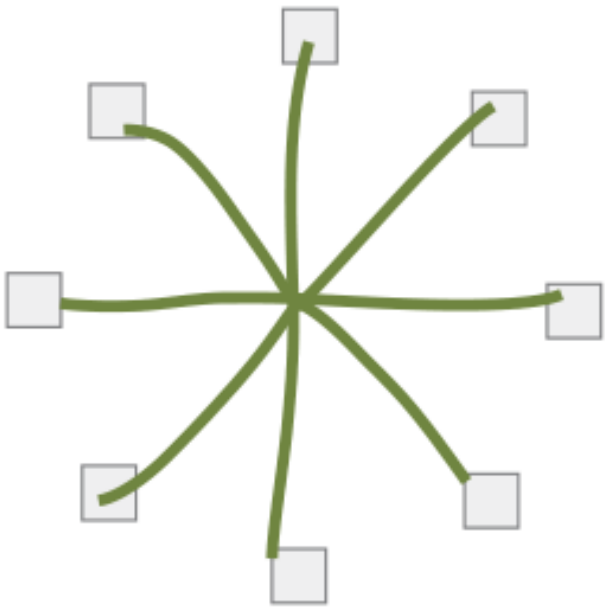
What changes with practice in the field?

FIELD ON · FIRST REACHES



Kandel 6e, Fig. 30–12D1, p. 731 · first reaches

FIELD ON · AFTER PRACTICE

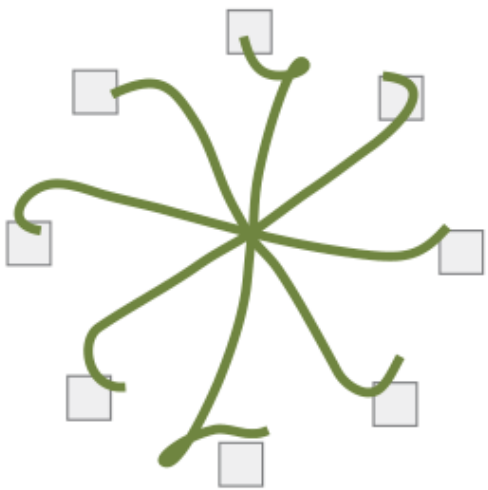


Kandel 6e, Fig. 30–12D2, p. 731 · later reaches

> Path improvement while the perturbing field remains supports compensation.

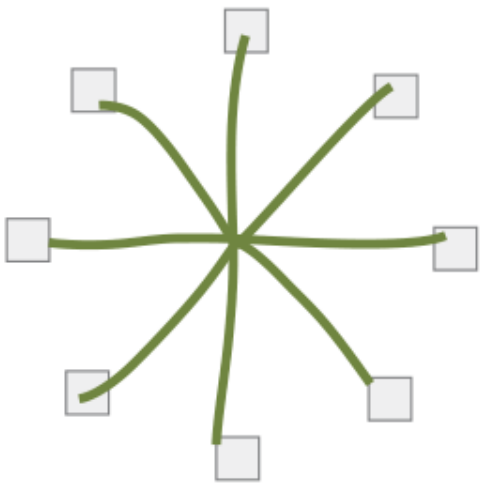
What happens when the field is removed?

FIELD ON · FIRST



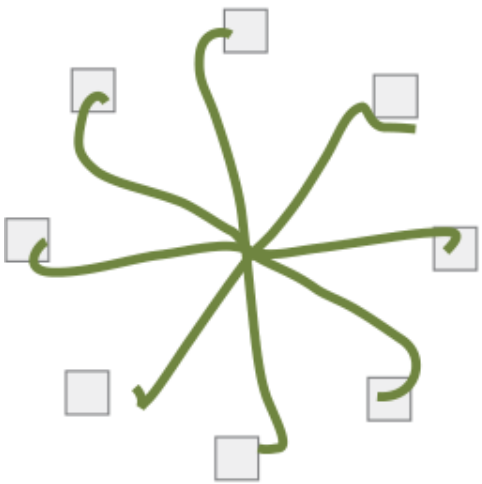
Kandel 6e, Fig. 30–12D1, p. 731

FIELD ON · LATER



Kandel 6e, Fig. 30–12D2, p. 731

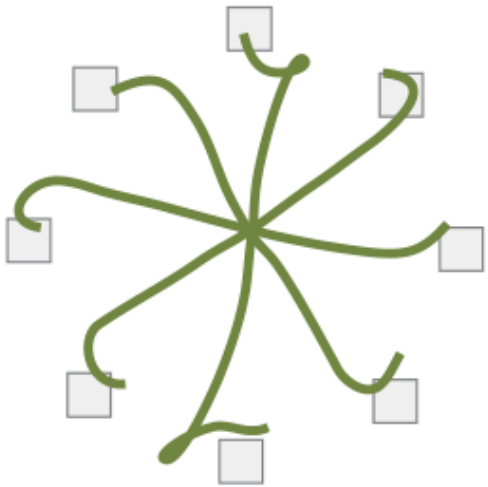
FIELD OFF



Kandel 6e, Fig. 30–12D3, p. 731

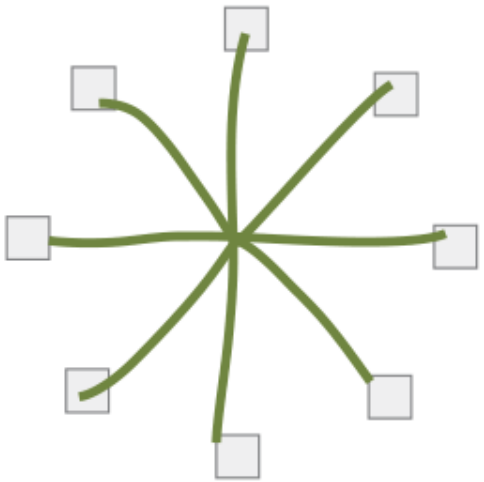
What happens when the field is removed?

FIELD ON · FIRST



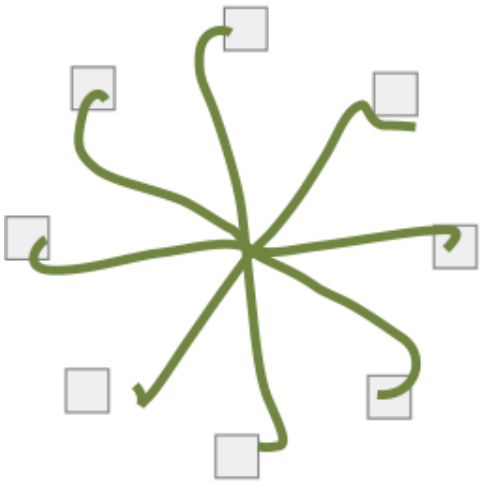
Kandel 6e, Fig. 30–12D1, p. 731

FIELD ON · LATER



Kandel 6e, Fig. 30–12D2, p. 731

FIELD OFF



Kandel 6e, Fig. 30–12D3, p. 731

> The deviation after removal is opposite the initial one: an aftereffect of learned compensation.

Prediction error and target error answer different questions

PREDICTION ERROR

Was the sensed consequence different from the expected one?

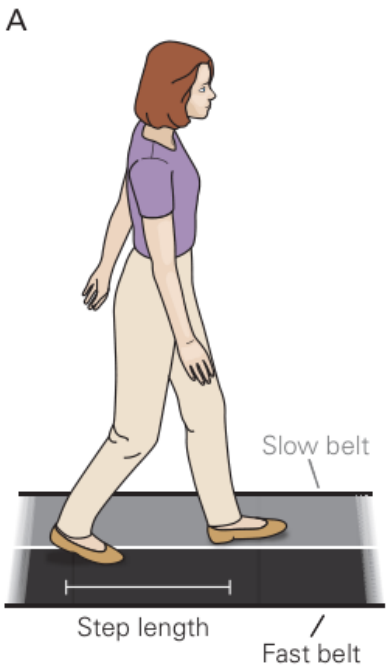
TARGET ERROR

Did the action miss its behavioral goal?

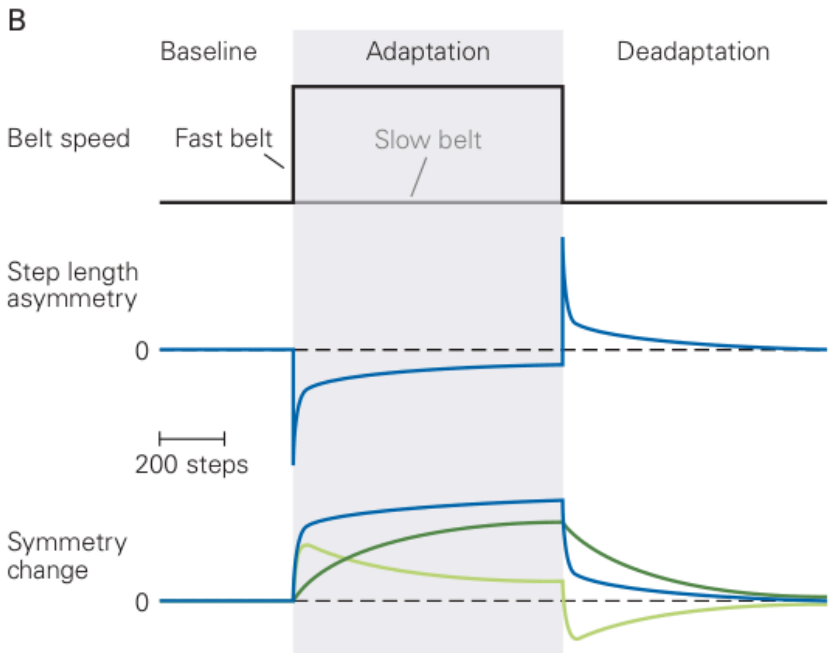
> A predicted miss can have target error with little surprise.

Kandel 6e, pp. 718, 730–731; Figs. 30–3 and 30–12 · conceptual contrast.

Split-belt walking changes step symmetry



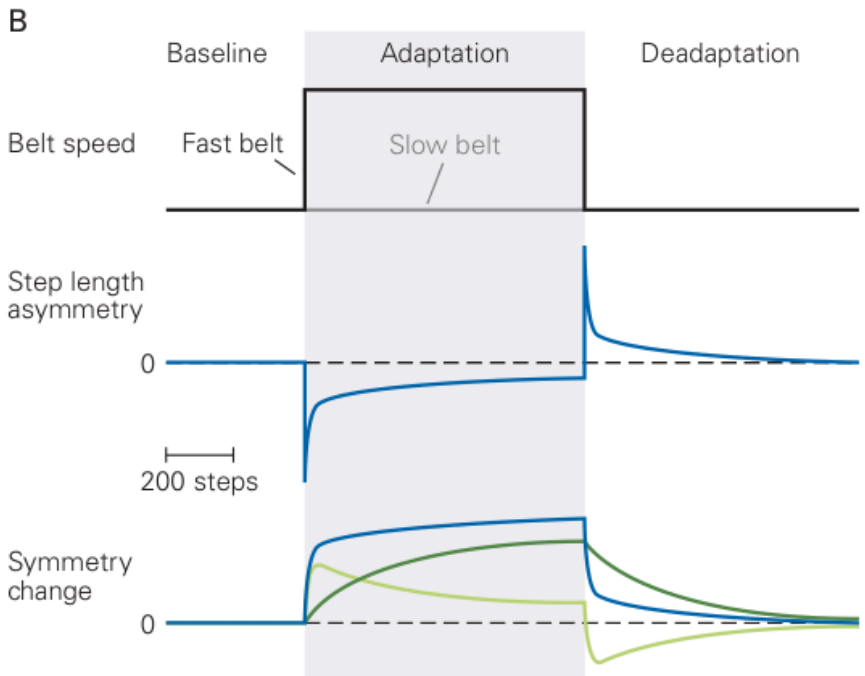
Kandel 6e, Fig. 30–13A, p. 732 · treadmill



Kandel 6e, Fig. 30–13B, p. 732 · illustrated time course and model

Step asymmetry appears, diminishes with practice, and reverses transiently at washout.

Fast and slow processes can fit the gait time course



Kandel 6e, Fig. 30–13B, p. 732 · two-rate model fit

> The curves summarize different adaptation rates; they are not recordings of two neuron populations.

Skill learning poses a wider credit-assignment problem

ADAPTATION

Compensate for an identifiable perturbation

SKILL ACQUISITION

Improve a policy when many choices contribute to delayed success

> Adaptation and skill learning overlap, but solve different parts of motor learning.

Kandel 6e, Ch. 30, pp. 732–735 · reading synthesis.

Common principles, different implementations

GAZE

Rapid orientation and stabilization

STANCE

Stability amid perturbation

GAIT

Rhythm, guidance, and adaptation

REACHING

Flexible, task-dependent goals

> Prediction, feedback, and learning recur across systems with partly distinct circuitry.

Kandel 6e, Ch. 30 and Figs. 30–1, 30–11, 30–13 · lecture synthesis.